

Consider a Ramsey-Cass-Koopmans model with the production function $Y = F(K, AL)$ and real interest rate r , household's utility function takes the form

$$U = \int_0^\infty e^{-\rho t} u(c(t)) \frac{L(t)}{H} dt$$

and there is CRRA (constant-relative-risk-aversion) utility

$$U(c(t)) = \frac{c(t)^{1-\theta}}{1-\theta}$$

(a) Construct Household budget constraint and show that there is no-ponzi-game condition.

(b) Solve Euler equation for household's maximization problem.

(c) Describe the behavior of economy in terms of the evolution of c and k .

(d) How does the competitive equilibrium maximize the welfare representative household?

(a) The representative household takes the paths of r and w as given. Its budget constraint is the present value of its life-time consumption cannot exceed its initial wealth plus the present value of its lifetime labor income.

Since the household has $L(t)/H$ members, its labor income at t is $w(t) L(t)/H$ and its consumption expenditures are $c(t) L(t)/H$. Its

initial wealth is $1/H$ of total wealth at time 0, or $k(0)/H$. The household's budget constraint is therefore -

$$\int_{t=0}^{\infty} e^{-R(t)} c(t) \frac{1}{H} dt \leq \frac{k(0)}{H} + \int_{t=0}^{\infty} e^{-R(t)} w(t) \frac{1}{H} dt \dots (1)$$

In general, it is not possible to find the integrals in this expression. Fortunately, we can express the budget constraint in terms of the limiting behavior of the household's capital holdings; and it is usually possible to describe the limiting behavior of the economy. To see how budget constraint can be rewritten in this way can be showed from rearranging eq(1)

$$\frac{k(0)}{H} + \int_{t=0}^{\infty} e^{-R(t)} [w(t) - c(t)] \frac{1}{H} dt \geq 0 \dots (11)$$

We can write the integral from $t=0$ to $t=\infty$ as a limit. Thus eq(11) is equivalent to

$$\lim_{S \rightarrow \infty} \left[\frac{k(0)}{H} + \int_{t=0}^S e^{-R(t)} [w(t) - c(t)] \frac{1}{H} dt \right] \geq 0 \dots (11')$$

The household's capital holdings at time S are

$$\frac{k(S)}{H} = e^{R(S)} \frac{k(0)}{H} + \int_{t=0}^S e^{R(S)-R(t)} [w(t) - c(t)] \frac{1}{H} dt \dots (12)$$

Here, $e^{R(S)} \frac{k(0)}{H}$ is the contribution of the household's initial wealth to its wealth at S . The household's saving at t is

$[W(t) - C(t)] \frac{1}{H}$, $e^{R(s) - R(t)}$ shows how the value of that saving changes from t to s . The expression in (iv) is $R(s)$ times the expression in brackets in (iii). Thus we can rewrite the budget constraint as simply

$$\lim_{s \rightarrow \infty} e^{-R(s)} \frac{K(s)}{H} \geq 0 \quad (v)$$

Expressed in this form, the budget constraint states that the present value of the household's asset holdings cannot be negative in the limit.

Eq (v) is known as the no-ponzi game condition. A ponzi game is a scheme in which someone issues debt and rolls it over forever. That is the issuer always obtains the funds to pay off debts when it comes due by issuing new debt. Such a scheme allows the issuer to have a present value of a lifetime consumption that exceeds the present value of his/her lifetime resources. By imposing eq (i) or (v) we are ruling out such scheme.

(b) The household's problem is to choose the path of $c(t)$ to maximize lifetime utility subject to budget constraint.

We can therefore use the objective function and budget constraint to set up Lagrangian:

$$\mathcal{L} = B \int_{t=0}^{\infty} e^{-\beta t} \frac{c(t)^{1-\theta}}{1-\theta} dt + \lambda \left[K(0) + \int_{t=0}^{\infty} e^{-R(t)} e^{(n+g)t} w(t) dt - \int_{t=0}^{\infty} e^{-R(t)} e^{(n+g)t} c(t) dt \right]$$

The household chooses c at point in time; that is, it chooses infinitely many $c(t)$'s. The first order condition for individual $c(t)$ is:

$$B e^{-\beta t} c(t)^{-\theta} = \lambda e^{-R(t)} e^{(n+g)t}$$

From this we can find:

$$\begin{aligned} \frac{\dot{c}(t)}{c(t)} &= \frac{r(t) - n - g - \beta}{\theta} \\ &= \frac{r(t) - \rho - \theta g}{\theta} \quad \dots (i) \end{aligned}$$

The second line uses the definition of β as $\rho - n - (1-\theta)g$. To interpret

(i), since $c(t)$ (consumption per worker) equals $\alpha(t) A(t)$, the growth rate of c is given by

$$\frac{\dot{c}(t)}{c(t)} = \frac{\dot{A}(t)}{A(t)} + \frac{\dot{\alpha}(t)}{\alpha(t)}$$

$$= g + \frac{r(t) - \rho - \theta g}{\theta}$$

$$= \frac{r(t) - \rho}{\theta}$$

where the second line uses eq(1). This condition states that consumption per worker is rising if the real return exceeds the rate at which the household discount future consumption and is falling if the reverse holds. The smaller is θ - the less marginal utility changes as consumption changes - the larger are the changes in consumption in response to differences between the real interest rate and the discount rate.

eq(1) is known as the Euler equation.

(c) The most convenient way to describe the behavior of the economy is in terms of the evolution of c and k .

For the dynamics of c we can consider the following equation:

$$\frac{\dot{c}(t)}{c(t)} = \frac{r(t) - \rho - \theta g}{\theta} = \frac{f'(k(t)) - \rho - \theta g}{\theta}$$

Thus $\dot{c}$ is zero when $f'(k)$ equals $\rho + \theta g$. Let k^* denote this level of k . When k exceeds k^* , $f'(k)$ is less than $\rho + \theta g$ and so $\dot{c}$ is negative; when k is less than k^* , $\dot{c}$ is positive.

For the dynamics of k , in the Solow model, $\dot{k}$ equals actual investment minus break-even investment. Since we are assuming that there is no depreciation, break-even investment is $(n+g)k$. Actual investment is output minus consumption, $f(k) - c$. Thus,

$$\dot{k}(t) = f(k(t)) - c(t) - (n+g)k(t)$$

For a given k , $\dot{k}$ is zero when consumption equals the difference between the actual output and breakeven investment line. When c exceeds the level that yields $\dot{k}=0$, k is falling, when c is less than this level, k is rising. The following fig combines these informations:

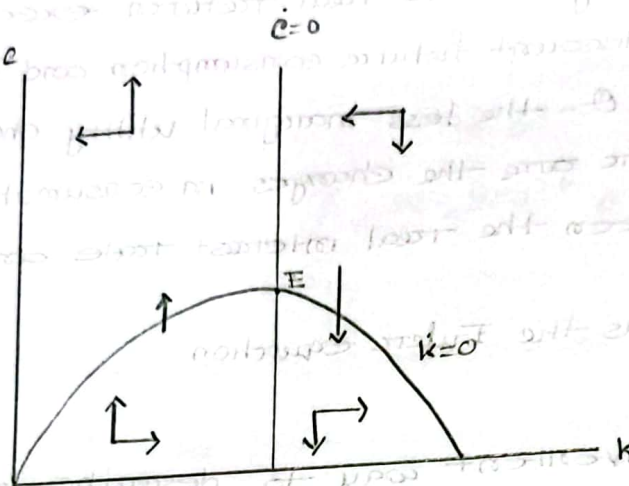


Fig: Dynamics of k and c

The arrows show the direction of motion of both k and c . To the left of the $\dot{c}=0$ locus and above the $\dot{k}=0$ locus, $\dot{c}$ is positive and $\dot{k}$ is negative. Thus c is rising and k is falling. On the $\dot{c}=0$ and $\dot{k}=0$ curves, only one of c and k is changing. On the $\dot{c}=0$ line above the $\dot{k}=0$ locus, c is constant and k is falling. Finally at point E both $\dot{c}$ and $\dot{k}$ are zero, thus there is no movement from this point.

(d) A natural question is whether the equilibrium of the economy represents a desirable outcome. The answer to this question is simple. The first welfare theorem from microeconomics tells us that if markets are competitive and complete and there are

no externalities, then the decentralized equilibrium is Pareto-efficient, that is, it is impossible to make anyone better off without making someone else worse off. Since the conditions of the first welfare theorem hold in the model, the equilibrium must be Pareto efficient. Since all households have same utility, this means that the decentralized equilibrium produces the highest possible utility among allocations that treat all households in the same way.

On the contrary, the solution to the social planner's problem is for the initial value of e to be given by the value of the saddle path, and for e and k to then move along the saddle path. That is competitive equilibrium maximizes the welfare of the representative household.

Compare The speed of convergence between Solow and REX model.

The speed of convergence in Solow model can be seen by the following equation:

$$\lambda \equiv -\frac{S k'(k)}{S k} = 1 - \alpha_k(k^*) (n+g+s)$$

This means k converges to its balanced growth path at rate $1 - \alpha_k(k^*) (n+g+s)$

For the speed of convergence to balance growth path

in RCK we can consider the following eq:

$$u_1 = \frac{1}{2} \left(\beta - \sqrt{\beta^2 + \frac{4}{\theta} \frac{1-\alpha}{\alpha} (\rho + \theta g) [\rho + \theta g - \alpha(\theta + g)]} \right)^{\frac{1}{2}}$$

This adjustment is quite rapid in this case; for comparison,

than is solow model for same values of α, n, g . For

$\alpha = 1/3$, $\rho = 4\%$, $n = 2\%$, $g = 1\%$ and $\theta = 1$; $u_1 = -5.4\%$ whereas

$\lambda = 2\%$. The reason for the difference is that the saving

rate is greater than s^* when k is less than k^* and less

than s^* when k is greater than k^* . In solow model

in contrast, s is constant by assumption.